THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum of the series $11 + 17 + 23 + \cdots + 113$ is: 1

(a) 1235 **(b)** 1116 **(c)** 2232 **(d)** 1003

2. In $\triangle ABC$ right-angled at B, $AB = 5$, $BC = 12$. If $BD \perp AC$, then AD (in cm) is: 1

(a) $\frac{5}{2}$ **(b)** $\frac{144}{13}$ **(c)** $\frac{60}{13}$ **(d)** $\frac{25}{13}$

3. The side of the square with diagonal endpoints $(1, 1)$ and $(7, 9)$ has squared length: 1

(a) 100 **(b)** 25 **(c)** 50 **(d)** 200

4. How many terms of the AP $18, 15, 12, 9, \ldots$ give sum 0? 1

(a) 13 **(b)** 14 **(c)** 26 **(d)** 12

5. A square has diagonal $7\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 14 **(b)** 49 **(c)** 196 **(d)** 98

6. The three points $(1, 1)$, $(9, 1)$ and $(4, 8)$ enclose area: 1

(a) 14 **(b)** 56 **(c)** 28 **(d)** 42

7. If the 3rd term of an AP is 8 and the 8th term is 3, then the 11th term is: 1

(a) 24 **(b)** 0 **(c)** 1 **(d)** 11

8. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 3 : 4$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 9: 1

(a) 16 **(b)** 58 **(c)** 40 **(d)** 49

9. The point on the y-axis equidistant from $(3, 1)$ and $(7, 7)$ has $y =$ 1

(a) $\frac{44}{3}$ **(b)** $-\frac{22}{3}$ **(c)** $\frac{22}{3}$ **(d)** 4

10. The sum of the first 15 terms of an AP is 465, and its first term is 10. The 15th term is: 1

(a) 72 **(b)** 62 **(c)** 21 **(d)** 52

11. In a right triangle with hypotenuse $c = 17$ and one leg $a = 8$, the area (sq. units) is: 1

(a) 60 **(b)** $\frac{255}{2}$ **(c)** 120 **(d)** 68

12. The area of the triangle formed by $(6, 0)$, $(0, 13)$ and the origin, doubled, is: 1

(a) 156 **(b)** 19 **(c)** 78 **(d)** 39

13. **Assertion (A):** The sequence $1, 4, 9, 16, \ldots$ is an AP. 1
Reason (R): In an AP, the difference between consecutive terms is constant.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A triangle with sides 5 cm, 6 cm and 7 cm is right-angled. 1
Reason (R): If in a triangle the square of one side equals the sum of the squares of the other two sides, the angle opposite the first side is a right angle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The point $(0, 5)$ lies on the x-axis. 1
Reason (R): The y-coordinate of every point on the x-axis is zero.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$ cm, $AB = 7$ cm and $BC = 21$ cm. Find DE. 2

17. Show that the points $A(1, 2)$, $B(4, 6)$, $C(9, 6)$, $D(6, 2)$ are the vertices of a rhombus. 2

18. If $S_n = 2n^2 + 5n$, find the first term and the common difference. 2

19. The areas of two similar triangles are 32 cm² and 162 cm². If a side of the first is 4 cm, find the corresponding side of the second. 2

20. Find the value of k for which the points $(-2, 3)$, $(0, 7)$ and $(3, k)$ are collinear. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The line joining $(3, 3)$ and $(12, 9)$ is trisected by points P and Q. Find their coordinates. 3

22. Find the sum of all integers between 20 and 300 that leave remainder 4 when divided by 9. 3

23. $\triangle ABC$ is right-angled at B with $AB = 5$ cm and $BC = 12$ cm; $BD \perp AC$. Find AD, DC and BD. 3

24. Find the values of y for which the distance between $P(17, y)$ and $Q(5, 1)$ is 13 units. 3

OR

In an AP, $S_7 = 98$ and $S_{12} = 228$. Find the AP. 3

25. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 5$ cm and $DC = 20$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

OR

Show that the points $A(2, 3)$, $B(5, 7)$, $C(10, 7)$, $D(7, 3)$ are the vertices of a rhombus. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Case study. On a town map drawn on a grid (1 unit = 100 m), the school is at $S(2, 3)$, the library at $L(8, 11)$ and the hospital at $H(2, 11)$. 5

(i) Find the coordinates of the mid-point of SL.

(ii) Find the distance SL in units.

(iii) Find the distance LH.

(iv) Show that the triangle SHL is right-angled and state the right-angle vertex.

27. Two poles of heights 45 m and 9 m stand on level ground 27 m apart. Find the distance between their tops. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(1, 5)$, $B(7, 3)$ and $C(10, 9)$. Find the fourth vertex D. 5

END OF QUESTION PAPER — Total: 50 Marks

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